The Besicovitch compression phenomenon and the Kakeya set conjecture

A report on Joshua Zahl's talk

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0 Introduction

This talk is about why the Kakeya conjecture in $\mathbb{R}^3$ matters, rather than about the proof of its 2025 resolution by Wang and Zahl. The aim is to state the problem clearly and to place it within harmonic analysis. Boards 1 and 2 give the conjecture together with its discretization into δ-tubes. Boards 3 through 12 then present three motivations for studying it: Fefferman's ball multiplier, Stein's Fourier restriction conjecture, and the free Schrödinger equation, all of which are tied to the Kakeya conjecture through the wave-packet dictionary. Board 13 and 14 states the main theorem of that resolution, phrased through the Convex Wolff Axiom.

1 Kakeya sets and the Kakeya conjecture

BOARD 1

Definition. A Besicovitch set is a compact $K \subset \mathbb{R}^n$ that contains a unit line segment pointing in **every** direction.

Besicovitch (1919). There exists a Besicovitch set $K \subset \mathbb{R}^2$ with $|K| = 0$.

Kakeya set conjecture. Every Besicovitch set in $\mathbb{R}^n$ has Minkowski and Hausdorff dimension n.

By Besicovitch's example, a compact set can contain a unit segment in every direction yet have Lebesgue measure 0, by a construction later known as the Perron tree. Since the measure can

collapse to zero, one must measure "size" with a finer instrument: dimension. The Kakeya conjecture asserts that however small the measure, the dimension of the set must be full, equal to n. One always has $\dim_H \le \dim_M$, so a full Hausdorff dimension forces a full Minkowski dimension; the Hausdorff statement is therefore the stronger one. Wang–Zahl prove both for $n = 3$.

2 Discretization of the Kakeya conjecture

BOARD 2

(Discretized) Kakeya conjecture. For all $\delta > 0$, let $\mathbb{T}$ be a set of δ-tubes in $\mathbb{R}^n$ pointing in δ-separated directions. Then

$$\Big|\bigcup_{T\in\mathbb{T}} T\Big| \;\ge\; \Big(\log\tfrac{1}{\delta}\Big)^{-C_n} \sum_{T\in\mathbb{T}} |T|,$$

and $\forall\,\varepsilon > 0$, $\exists\, c_\varepsilon > 0$ such that

$$\Big|\bigcup_{T\in\mathbb{T}} T\Big| \;\ge\; c_\varepsilon\,\delta^{\varepsilon} \sum_{T\in\mathbb{T}} |T|.$$

Here $\mathbb{T}$ is the set of tubes and T a single δ-tube: the δ-neighborhood of a unit segment, of volume $|T| \sim \delta^{n-1}$. Throughout, $|\cdot|$ denotes Lebesgue measure; "δ-separated directions" means the tubes' directions are pairwise at least δ apart; C_n is a constant depending only on n, and c_ε is a constant depending only on ε.

Full dimension as $\delta \to 0$ is equivalent to such a volume lower bound: with δ-separated directions $\mathbb{T}$ contains $\#T \sim \delta^{-(n-1)}$ tubes and $\sum_{T\in\mathbb{T}} |T| \sim 1$, so the conjecture forces $\big|\bigcup_{T\in\mathbb{T}} T\big|$ to be comparably large.

Both inequalities bound the same $\big|\bigcup_{T\in\mathbb{T}} T\big|$ from below, so the stronger one is whichever carries the larger prefactor. Set $L := \log\frac{1}{\delta}$, so that $L \to +\infty$ as $\delta \to 0$. The factor $(\log\frac{1}{\delta})^{-C_n} = L^{-C_n}$ decays only polynomially in L (polylogarithmically in $1/\delta$), whereas $\delta^{\varepsilon} = e^{-\varepsilon L}$ decays exponentially in L. Since exponential decay outpaces any polynomial,

$$\lim_{\delta\to 0} \frac{(\log\frac{1}{\delta})^{-C_n}}{\delta^{\varepsilon}} = \lim_{L\to\infty} \frac{e^{\varepsilon L}}{L^{C_n}} = +\infty,$$

so $(\log\frac{1}{\delta})^{-C_n} \ge c_\varepsilon\,\delta^{\varepsilon}$ once δ is small. Chaining the two displays, the first inequality implies the second but not conversely, so it is the larger, more precise lower bound.

Córdoba proves the first inequality for $n = 2$ with $C_2 = 1$; in dimension 3, Wang and Zahl prove the second inequality, the slightly weaker statement in which $(\log\frac{1}{\delta})^{-C_n}$ is replaced by $c_\varepsilon\delta^{\varepsilon}$.

3 Motivation I: Fefferman's ball multiplier example

BOARD 3

Fefferman's ball multiplier example. For $f : \mathbb{R}^2 \to \mathbb{C}$,

$$Bf = \big(\hat{f}\cdot\chi_{B(0,1)}\big)^{\vee}.$$

Fefferman (1971). For all $p \neq 2$,

$$\|B\|_{L^p(\mathbb{R}^2)\to L^p(\mathbb{R}^2)} = \infty.$$

BOARD 8



$\chi_{B(0,1)}$ (sharp) smoothed (Bochner–Riesz)

The multiplier as a function of frequency: the sharp indicator $\chi_{B(0,1)}$ (left) versus its Bochner–Riesz smoothing $\big(\max(0,\,1-|\xi|^2)\big)^{\alpha}$ (right). The jump across $|\xi| = 1$ is the sharp edge that obstructs L^p boundedness; tapering it smoothly to 0 removes the obstruction once α is large.

B keeps only the frequencies inside the unit ball. On L^2 it is trivially bounded, since the Fourier transform is an isometry and the truncation only lowers the norm, and before 1971 it was expected to stay bounded for p in some range around 2 as well. Fefferman's 1971 theorem was the surprise that it is not: in $\mathbb{R}^n$, $n \geq 2$, the curvature of the sphere forces boundedness at $p = 2$ alone, and the proof runs through Besicovitch sets. This is the first reason the conjecture matters to analysts: a basic question about the Fourier transform in $\mathbb{R}^n$ turns out to hinge on Besicovitch compression, so even someone indifferent to Kakeya sets must reckon with them.

Determining the smallest α for which the smoothed multiplier $\big(\max(0,\,1-|\xi|^2)\big)^{\alpha}$ stays bounded is the famous Bochner–Riesz conjecture, settled in the plane and open from $\mathbb{R}^3$ on. It rests on the same compression: a Besicovitch set with stronger compression, run through Fefferman's construction, would break Bochner–Riesz the more violently, so the limit the Kakeya conjecture places on compression is what bounds how far Bochner–Riesz can fail (§4).

4 Motivation I (continued): unboundedness of the ball multiplier

BOARD 4

Fix $p > 2$ and $C \gg 1$. We want $f \in L^p(\mathbb{R}^2)$ with $\|Bf\|_p > C\|f\|_p$. If $\|B\|_{L^p\to L^p} \leq C$, then the following is true. Let $N \geq 1$, let $F = (f_1, \ldots, f_N)$ with each $f_i : \mathbb{R}^2 \to \mathbb{C}$, let $v_1, \ldots, v_N$ be unit vectors in $\mathbb{R}^2$, and set

$$TF = (H_1 f_1, \ldots, H_N f_N), \qquad H_i = \text{Hilbert transform in direction } v_i.$$

BOARDS 5 & 7

Then the vector-valued inequality holds:

$$\big\| \|TF\|_{\ell^2} \big\|_{L^p} \;\leq\; C \big\| \|F\|_{\ell^2} \big\|_{L^p},$$

that is,

$$\int \Big(\sum_{i=1}^{N} |H_i f_i|^2 \Big)^{p/2} \;\leq\; C^p \int \Big(\sum_{i=1}^{N} |f_i|^2 \Big)^{p/2}.$$

Board 7 annotates: the right-hand integral is ~ 1, so the right side is $\sim C^p$ (a fixed constant), while the left-hand integral grows without bound in N and eventually exceeds it, so the inequality fails and B is unbounded.

BOARD 6

Take $f_i = \chi_{R_i}$, with v_i the direction of R_i: these are N rectangles of length 1 and thickness $\delta = \frac{1}{N}$ in separated directions, essentially disjoint (so $\left|\bigcup_i R_i\right| \sim 1$). Each H_i moves its rectangle along v_i: $|H_i f_i| \sim 1$ on a translate $\tilde{R}_i$, and these translates $\tilde{R}_i$ form a collection whose union is very small (a Perron tree).

Suppose $\|B\|_{L^p \to L^p} \le C$. The unit sphere presents every direction as a boundary normal, so localized near each boundary point B reduces to the Hilbert transform in that direction, and Khintchine randomization assembles these into the vector-valued inequality on boards 5/7. Test it on board 6's rectangles $f_i = \chi_{R_i}$: thin $1 \times \frac{1}{N}$ rectangles pointing in separated directions, taken pairwise disjoint, so the right-hand integral is ~ 1 and the right side is $\sim C^p$. The Hilbert transform H_i smears χ_{R_i} along its direction, leaving $|H_i f_i| \sim 1$ on a unit translate $\tilde{R}_i$; by the Besicovitch/Perron construction these translates overlap into a set of measure $\varepsilon \to 0$ even though $\sum_i |\tilde{R}_i| \sim 1$. For $p > 2$ that concentration drives the left-hand integral to infinity with N, so it eventually overtakes C^p and the inequality fails, a contradiction giving $\|B\|_{L^p \to L^p} = \infty$. This is exactly the Besicovitch phenomenon: the sphere's curvature brings in every direction at once, so bounding B would mean controlling directional Hilbert transforms in all directions simultaneously, the Kakeya problem, which is what ties the ball-multiplier and Bochner–Riesz problems to the Kakeya conjecture.

5 Motivation II: Stein's Fourier restriction conjecture

BOARD 9

Stein's restriction conjecture ($\mathbb{R}^3$). For $f : \mathbb{R}^2 \to \mathbb{C}$, define the extension operator over the paraboloid,

$$Ef(x) = \int e^{\,2\pi i\,(x_1\xi_1 + x_2\xi_2 + x_3(\xi_1^2+\xi_2^2))}\, f(\xi)\, d\xi,$$

equivalently $Ef = (g\, d\sigma)^\vee$ with $g(\xi_1, \xi_2, \xi_1^2 + \xi_2^2) = f(\xi_1, \xi_2)$ on the paraboloid $P = \{\xi \in \mathbb{R}^3 : \xi_3 = \xi_1^2 + \xi_2^2\}$.

Here $x = (x_1, x_2, x_3) \in \mathbb{R}^3$ is the physical space variable (so Ef is a function on $\mathbb{R}^3$); the third coordinate x_3 is sometimes written t, as it plays the role of time when E reappears as the Schrödinger propagator in §6. The pair $(\xi_1, \xi_2) \in \mathbb{R}^2$ are the two base frequencies that f depends on. Frequency space is $\mathbb{R}^3$ with coordinates (ξ_1, ξ_2, ξ_3), and the paraboloid P is where the third coordinate is pinned to the first two by $\xi_3 = \xi_1^2 + \xi_2^2$, the graph of $(\xi_1, \xi_2) \mapsto \xi_1^2 + \xi_2^2$, a curved surface. The function g is f lifted onto P: projecting the (ξ_1, ξ_2)-plane up to the paraboloid, $g(\xi_1, \xi_2, \xi_1^2 + \xi_2^2) = f(\xi_1, \xi_2)$ places the values of f on the surface. The measure $d\sigma$ on P is defined by projection: the σ-measure of a subset of P is the two-dimensional Lebesgue measure of its projection back down to the (ξ_1, ξ_2)-plane, so in the graph parametrization it is simply $d\sigma = d\xi_1\, d\xi_2$, flat Lebesgue measure pulled up onto P, *not* the curved surface-area element. With this normalization the two definitions coincide *verbatim*. Unwinding $(g\, d\sigma)^\vee$ in the graph

parametrization,

$$(g\,d\sigma)^\vee(x) = \int_{\mathbb{R}^2} e^{\,2\pi i\, x\cdot(\xi_1,\xi_2,\,\xi_1^2+\xi_2^2)}\, g(\xi_1,\xi_2,\xi_1^2+\xi_2^2)\,d\xi_1\,d\xi_2$$
$$= \int_{\mathbb{R}^2} e^{\,2\pi i\,(x_1\xi_1+x_2\xi_2+x_3(\xi_1^2+\xi_2^2))}\, f(\xi)\,d\xi \;=\; Ef(x),$$

where the second equality inserts $g(\xi_1,\xi_2,\xi_1^2+\xi_2^2) = f(\xi_1,\xi_2)$ and $d\sigma = d\xi_1\,d\xi_2$. Had $d\sigma$ instead been the surface-area element $\sqrt{1+4\xi_1^2+4\xi_2^2}\,d\xi_1\,d\xi_2$, that Jacobian would survive and the two forms would no longer agree; with the projected measure there is nothing deep in the equivalence: it is the same object. This is the sense in which E *extends* a function on $\mathbb{R}^2$ to one on $\mathbb{R}^3$, by riding it on the paraboloid.

BOARD 10

Question. For which p does

$$\|Ef\|_{L^p(\mathbb{R}^3)} \le C_p\,\|f\|_{L^p(\mathbb{R}^2)}$$

hold? **Conjecture:** $p > 3$, i.e. $p > \dfrac{2n}{n-1}$.

The restriction conjecture asks for which p this bound holds. Because $d\sigma$ is the lifted measure, the right-hand side $\|f\|_{L^p(\mathbb{R}^2)}$ equals $\|g\|_{L^p(d\sigma)}$, so the estimate is the standard extension form $\|(g\,d\sigma)^\vee\|_{L^p(\mathbb{R}^3)} \le C_p\,\|g\|_{L^p(d\sigma)}$. The conjectured threshold is $p > \frac{2n}{n-1}$, thus $p > 4$ in $\mathbb{R}^2$ and $p > 3$ in $\mathbb{R}^3$. In $\mathbb{R}^2$ it is a theorem (Fefferman and Zygmund); from $n = 3$ on it is open, with only partial results over restricted ranges of p, so $\mathbb{R}^3$ is the first unknown case. The curvature of P is what makes a nontrivial bound possible (a flat surface would give none), and it is the same curvature that ties restriction to Kakeya (§7).

6 Motivation III: the free Schrödinger equation

BOARD 11

Free Schrödinger equation in 2+1 dimensions. For $u(x,t)$,

$$-2\pi i\,\partial_t u + \Delta_x u = 0, \qquad u(x,0) = u_0(x).$$

Let $F = \hat{u}_0$; then $u = EF$.

Derivation. Take the space–time Fourier transform, with ξ dual to x and τ dual to t; then $\partial_t \mapsto 2\pi i\tau$ and $\Delta_x \mapsto -4\pi^2|\xi|^2$, so the equation becomes

$$(-2\pi i)(2\pi i\tau)\,\hat{u} - 4\pi^2|\xi|^2\,\hat{u} = 0 \implies 4\pi^2\big(\tau - |\xi|^2\big)\,\hat{u} = 0$$
$$\implies \big(\tau - |\xi|^2\big)\,\hat{u} = 0.$$

Hence $\hat{u}$ is supported on the paraboloid $\tau = |\xi|^2$: the solution's space–time frequencies all lie on it, which is precisely the assertion $u = EF$.

So the propagator *is* the extension operator, and the restriction question reads as a dispersion estimate (if the initial datum's transform lies in L^p, is the space-time solution in L^p?), with smaller admissible p meaning faster spreading of energy. The same correspondence attaches other surfaces to other equations: the sphere to the Helmholtz equation, the cone to the wave

equation. So L^p extension bounds are Strichartz estimates, $t \to 0$ convergence is Carleson's problem, and the like; Kakeya lies beneath all of them.

7 How the restriction conjecture implies the Kakeya conjecture

BOARD 12

Left: the paraboloid, thickened to its R^{-1}-neighborhood, is cut into caps (tangential size $R^{-1/2}$, thickness R^{-1}); writing $g = \sum_i g_i$ gives $Eg = \sum_i Eg_i$. Right: inside a ball of radius R, the dual tubes (length R, thickness $R^{1/2}$, each pointing along the normal to its cap) cross one another, and on each tube the wave packet has roughly constant magnitude, $|Eg_i| \sim$ const.

First localize to a ball of radius R, accepting an R^ε loss; proving that weaker bound for every $\varepsilon > 0$ recovers the conjecture over $\mathbb{R}^n$ on the same range of p. At this scale the uncertainty principle blurs P to its R^{-1}-neighborhood, whose largest convex pieces are caps of tangential size $R^{-1/2}$ and thickness R^{-1}. Decomposing $g = \sum_i g_i$ over the caps, linearity gives $Eg = \sum_i Eg_i$, and each Eg_i (an inverse Fourier transform of a function on a convex cap) has magnitude roughly constant on the dual plank, of size $R^{1/2} \times R^{1/2} \times R$ and pointing along the cap's normal. Different caps have different normals, so the planks point in different directions; rescaling B_R to the unit ball turns them into δ-tubes ($\delta = R^{-1/2}$) in δ-separated directions, exactly a Kakeya set.

The bound $\|Eg\|_p \lesssim \|g\|_{L^p(d\sigma)}$ now demands that summing these wave packets barely raises the norm, as though the pieces were orthogonal, and this splits in two. Were the tubes disjoint the pieces would be genuinely orthogonal; what holds instead is that they *barely overlap*, and precisely how little they overlap is the Kakeya statement, the geometric content. Where tubes do overlap, the Eg_i carry complex phases that may reinforce or cancel, and the extra force of restriction is that these phases enjoy square-root (random-sign) cancellation. Hence

$$\text{restriction conjecture} \implies \text{Kakeya conjecture} \qquad \text{(Bourgain / Córdoba)}:$$

Kakeya is the geometric skeleton (tube overlap consistent with orthogonality), and restriction layers the oscillatory cancellation on top. This is why Kakeya is a genuine obstruction beneath restriction, and why settling Kakeya in $\mathbb{R}^3$ does not by itself settle restriction there.

8 The Convex Wolff Axiom (CWA) and the main theorem

Zahl closes the talk by stating the main theorem on the Kakeya conjecture in $\mathbb{R}^3$. It is the volume inequality of §2 once more, the same $\left|\bigcup T\right| \geq c_\varepsilon \delta^\varepsilon \sum |T|$, but now for a larger class of tube families than the δ-separated ones: those satisfying the Convex Wolff Axiom.

BOARD 13

Main theorem ($\mathbb{R}^3$). For every $\varepsilon > 0$ there is $c_\varepsilon > 0$ such that for all $\delta > 0$ and every set $\mathbb{T}$ of δ-tubes in $\mathbb{R}^3$ satisfying CWA,

$$\Big|\bigcup_{T\in\mathbb{T}} T\Big| \;\geq\; c_\varepsilon\, \delta^\varepsilon \sum_{T\in\mathbb{T}} |T|.$$

BOARD 14

Definition (CWA). $\mathbb{T}$ satisfies the Convex Wolff Axiom if, for every convex set $W \subset \mathbb{R}^3$,

$$\#\{T \in \mathbb{T} : T \subset W\} \le C\,|W|/\delta^2, \qquad \sum_{\substack{T \in \mathbb{T} \\ T \subset W}} |T| \le C\,|W|.$$

Here $|W|/\delta^2$ is the number of disjoint δ-tubes that fit inside W, so CWA caps the tube count in every convex window at that of a packing, which is exactly the way the bound can break. If many tubes were crammed into a single convex W, then $\left|\bigcup T\right|$ could be as small as $|W|$ while $\sum |T|$ stayed large; CWA forbids precisely this crowding.

A δ-separated family satisfies CWA automatically, so the theorem is strictly stronger than the Kakeya conjecture, which, as the closing explains, is exactly what makes CWA the right hypothesis to prove.

9 Closing

Zahl closes with the lesson the theorem carries. For decades the obstacle had seemed to lie in the geometry of δ-separated directions, but the realization behind the proof in $\mathbb{R}^3$ is that this was the wrong notion. The right question is what kind of non-concentration prevents the volume bound from failing, and the answer is convexity rather than separation. A family of tubes can defeat the bound only by crowding into a convex set, and in $\mathbb{R}^3$ that is the only obstruction, up to the δ^{ε} loss. The result is therefore best understood as a theorem in convex geometry: extremely eccentric convex sets, of dimensions $1 \times \delta \times \delta$, are almost disjoint from one another unless they are trapped inside a common convex set. The contrast with balls is the telling one. For balls a Vitali covering is available, because a bounded dilate of the larger of two overlapping balls already contains the smaller one, whereas covering a single $1 \times \delta$ slab by a dilate of a transverse slab would require a dilate of factor δ^{-1}. No covering lemma rescues the naive argument, and this is why the estimate has to be earned. It is also the origin of the paper's title, *Volume estimates for unions of convex sets*.

Convexity is also what survives the machinery of the proof. The argument runs on repeated rescalings and an induction on scales, and the convex statement is preserved under all of these operations, whereas δ-separation is not. The theorem is, moreover, special to $\mathbb{R}^3$. The statement continues to make sense when $n \ge 4$, but in those dimensions it is false: a quadratic hypersurface can contain a great many lines, and once these are thickened it contains a great many δ-tubes, which gives families that satisfy the Convex Wolff Axiom and yet concentrate. This failure is new to four dimensions and above. Reasonable questions remain in higher dimensions, but they are difficult, and the three-dimensional argument does not reach them. The proof itself, which the talk did not cover, is the subject of the paper mentioned above.

End of report · reconstructed from fourteen blackboard photographs.